Summary of Computer Vision Laboratory Project

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Neural Network Based Handwritten Digit Classifier

The overall goal of this system is to build and demonstrate a robust digit recognition model using deep learning techniques, specifically neural networks. These neural networks utilize dense layers and activation functions to learn complex patterns in the data. The system is trained on the **MNIST** dataset and can accurately predict the digits in new handwritten input from users. Implementation was done with **Python3**, **TensorFlow** and **Keras, NumPy**, **Matplotlib** etc. libraries were used.

Model Building

Determining of architecture, activation functions (ReLU), optimizers (SGD) and loss functions

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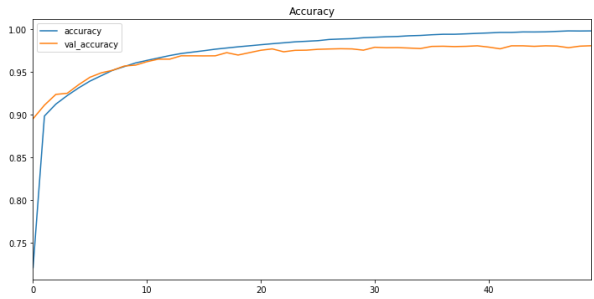
Model Training

Process of updating the model's parameters (weight and biases)
MNIST dataset is used

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Model Evaluation

Assessment of the trained model performance; metrics such as accuracy, precision, recall



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Prediction

Using trained model for recognizing handwritten digits



Input Digit Predicted Digit